Wrangling Data

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Remember that you can either fill in the "data_wrangling.qmd" file or follow along with the "data_wrangling_key.qmd" file.

Load Packages

The packages we need for our explorations today (`readr` for reading in data, `ggplot2` for graphing data, and `dplyr` for wrangling/summarizing the data) are part of a popular suite of packages called the `tidyverse`.

```
library(tidyverse)
library(ggrepel)
```

Data Background

We will return to the same dataset we saw last time. Here's the background and description of the variables.

In 2013, the government decided to make data about colleges more accessible so that students and parents could more easily compare schools. These data are called the "College Scorecard" data and the 2024 dataset contains 3,305 variables on 6,484 universities in the US!

I have filtered that 2024 dataset to only include schools which confer majority baccalaureate degrees and where the majority of those degrees are in the arts and sciences based on the Carnegie Classification system. In other words, I filtered the data down to the schools which are "similar" to Bucknell (including Bucknell itself) and picked out some variables for us to explore.

Data Dictionary

Below are the code names and descriptions of the variables in our dataset.

- `UNITID`: Unique identifier
- `INSTNM`: Name of institution
- `CITY`: City
- `STABBR`: State
- `HIGHDEG`: Highest degree awarded (0 = Non-degree grants, 1 = Certificate degree, 2 = Associate degree, 3 = Bachelor's degree, 4 = Graduate degree)
- `PREDDEG`: Predominant undergraduate degree awarded (0 = Not classified, 1 = Predominantly certificate-degree granting, 2 = Predominantly associate's-degree granting, 3 = Predominantly bachelor's-degree granting, 4 = Entirely graduate-degree granting)
- `CONTROL`: Ownership (1 = Public, 2 = Private non-profit, 3 = Private for-profit)
- `HBCU`: Flag for Historically Black College and University
- `TUITFTE`: Net tuition revenue per full-time equivalent student
- `AVGFACSAL`: Average faculty salary
- `ADM_RATE`: Admission rate
- `SATVR75`: 75th percentile of SAT scores at the institution (critical reading)
- `SATMT75`: 75th percentile of SAT scores at the institution (math)
- `ACTCM75`: 75th percentile of the ACT cumulative score
- `COSTT4_A`: The average annual total cost of attendance, including tuition and fees, books and supplies, and living expenses for all full-time, first-time, degree/certificate-seeking undergraduates who receive Title IV aid.
- `NPT4_PRIV`: The average annual total cost of attendance, including tuition and fees, books and supplies, and living expenses, minus the average grant/scholarship aid
- `UGDS`: Enrollment of undergraduate certificate/degree-seeking students
- `UG25ABV`: Percentage of undergraduates aged 25 and above
- `PCTFLOAN_DCS`: Percentage of degree/certificate-seeking undergraduate students awarded a federal loan
- `PCTPELL_DCS`: Percentage of degree/certificate-seeking undergraduate students awarded a Pell Grant
- `DEBT_MDN`: The median original amount of the loan principal upon entering repayment

- `C100_4`: Completion rate for first-time, full-time students at four-year institutions (100% of expected time to completion)
- `RET_FT4`: First-time, full-time student retention rate at four-year institutions
- `MD_EARN_WNE_5YR`: Median earnings of graduates working and not enrolled 5 years after completing

Load the Data

Run the following code to load and inspect the data.

```
# Load the data
colleges <- read_csv("data/ccbasic21.csv")


# Wrangling data (Will talk through soon!)
colleges <- colleges %>%
  mutate(Location = if_else(STABBR == "PA", "PA", "Not PA"),
         CONTROL_CAT = case_match(CONTROL,
                                  1 ~ "Public",
                                  2 ~ "Private non-profit"),
         HIGHDEG_CAT = case_match(HIGHDEG,
                                  3 ~ "Bachelor's degree",
                                  4 ~ "Graduate degree"))
```

What is data wrangling??

Data wrangling = any transformations done on the data.

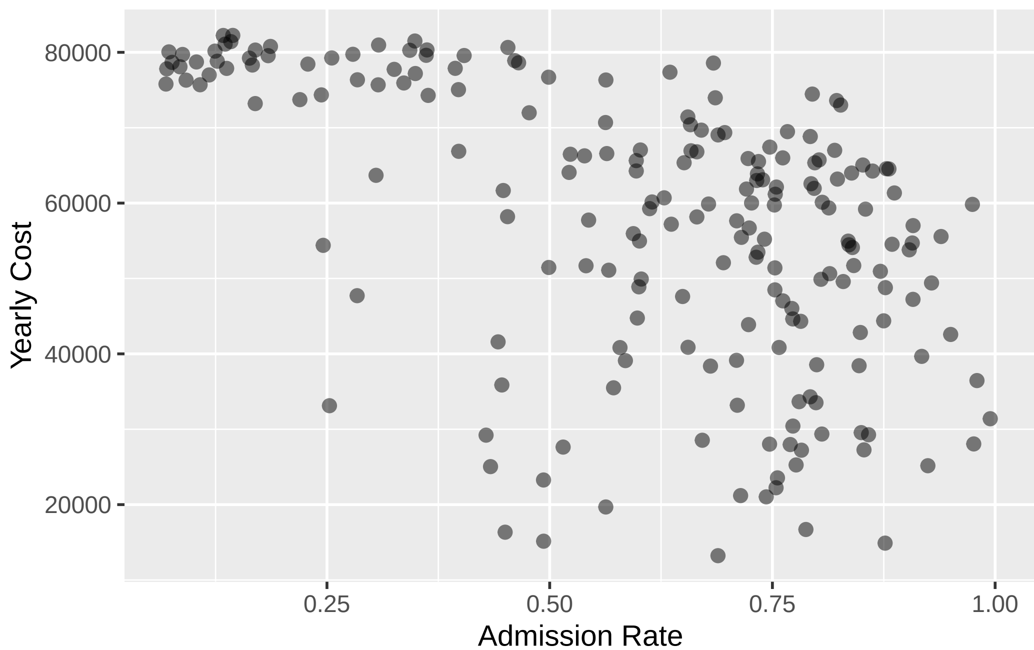
Examples:

- Summarizing the data by computing the mean of a variable.
- Counting the number of observations in the categories of a set of variables. (Excel users: Think pivot tables.)
- Dropping rows of the dataset that have missing values.
- Filtering down to just a subset of the data.
- Collapsing a categorical variable into fewer categories.
- Fixing how `R` stores a variable.
- Sorting the data by one of the variables.
- Joining multiple datasets together.
  - Won't see joins today but can learn about them here

Motivating Examples: Sprucing up our Graphs

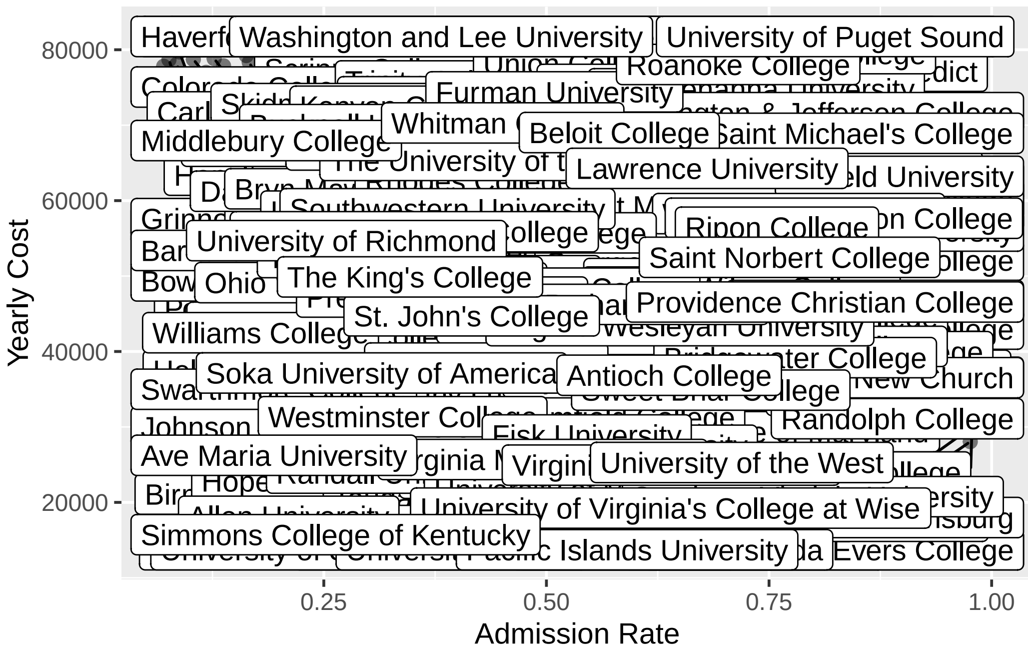
How do I add Bucknell to my graph?

```
ggplot(data = colleges,
       mapping = aes(x = ADM_RATE,
                     y = COSTT4_A)) +
  geom_point(size = 2, alpha = .5) +
  labs(x = "Admission Rate", y = "Yearly Cost")
```



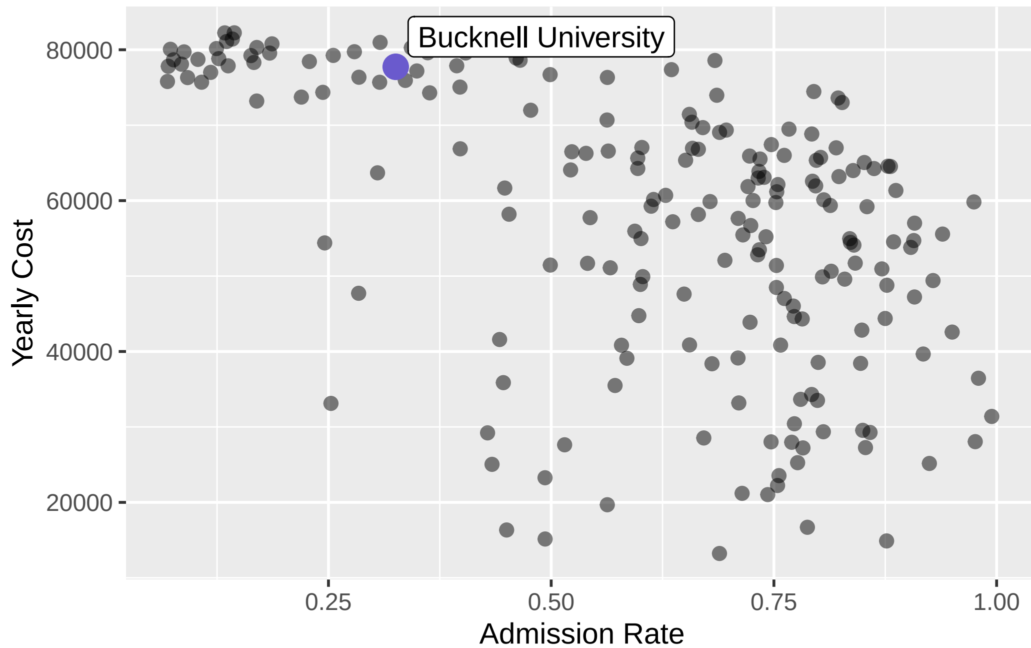
Option 1: Label all the schools.

```
ggplot(data = colleges,
       mapping = aes(x = ADM_RATE,
                     y = COSTT4_A)) +
  geom_point(size = 2, alpha = .5) +
  labs(x = "Admission Rate", y = "Yearly Cost") +
  geom_label_repel(mapping = aes(label = INSTNM), max.overlaps = 150)
```



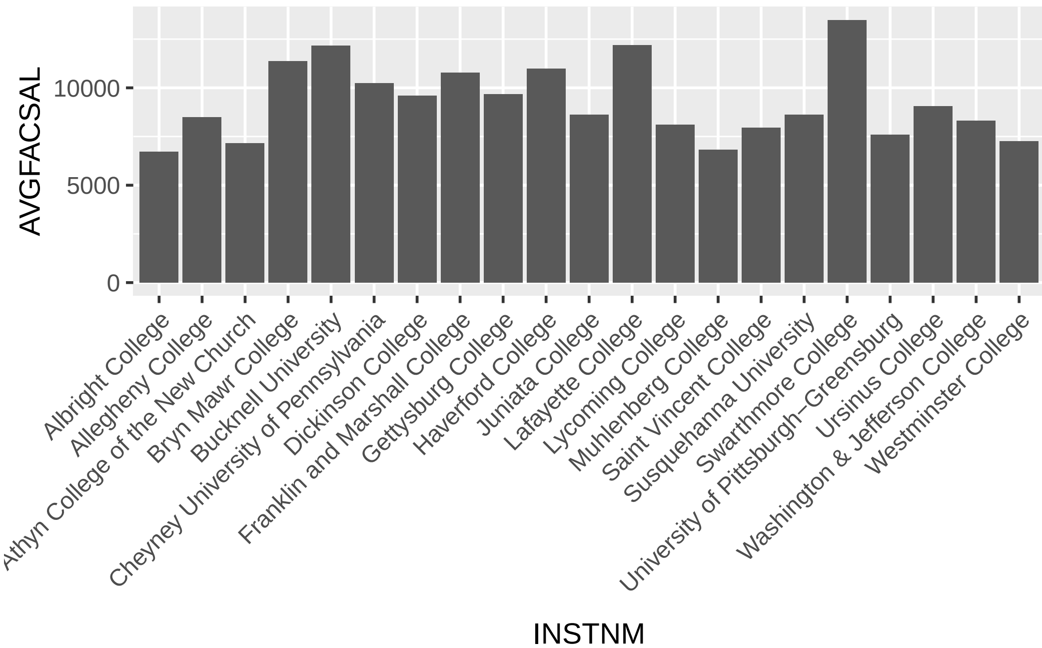
Option 2: Create a new dataset that contains only Bucknell and then add that to the graph.

```
# Create a dataset that just has Bucknell's info
Bucknell <- filter(colleges, INSTNM == "Bucknell University")

# Just label Bucknell
ggplot(data = colleges,
       mapping = aes(x = ADM_RATE,
                     y = COSTT4_A)) +
  geom_point(size = 2, alpha = .5) +
  labs(x = "Admission Rate", y = "Yearly Cost") +
  geom_label_repel(data = Bucknell, mapping = aes(label = INSTNM), max.overlaps = 20) +
  geom_point(data = Bucknell, size = 4, color = "slateblue")
```



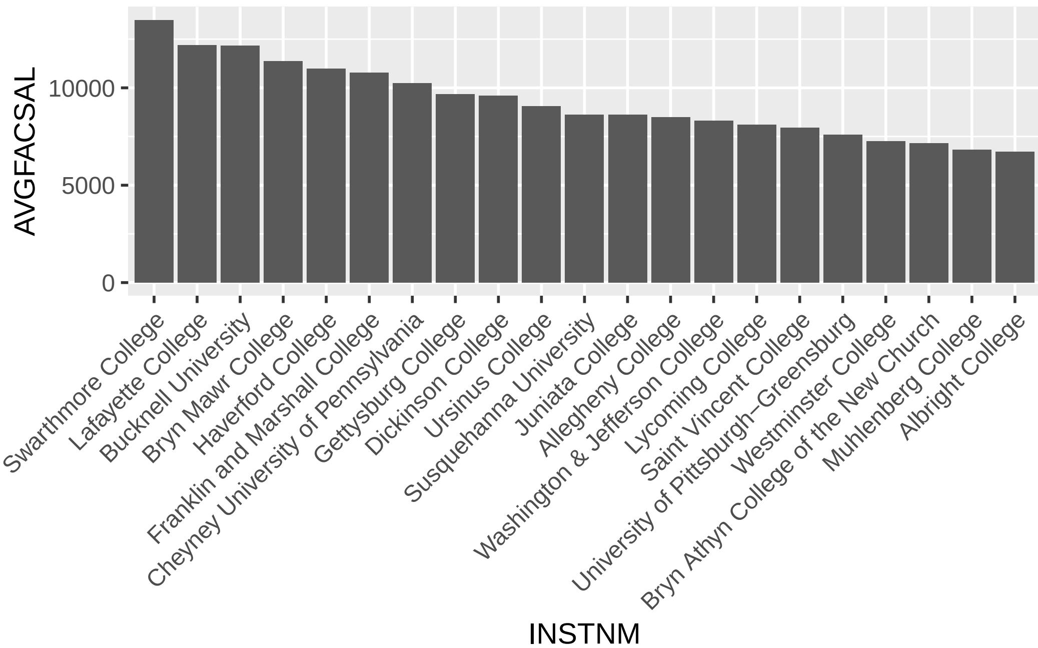
How do I reorder the bars of my bar graph?

```
# Let's focus on PA schools
pa_schools <- filter(colleges, STABBR == "PA")

# Look at average faculty salaries by school
ggplot(data = pa_schools, mapping = aes(x = INSTNM,
                                        y = AVGFACSAL)) +
  geom_col() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



What is the current order for `INSTNM`? What is a better order for the bars?

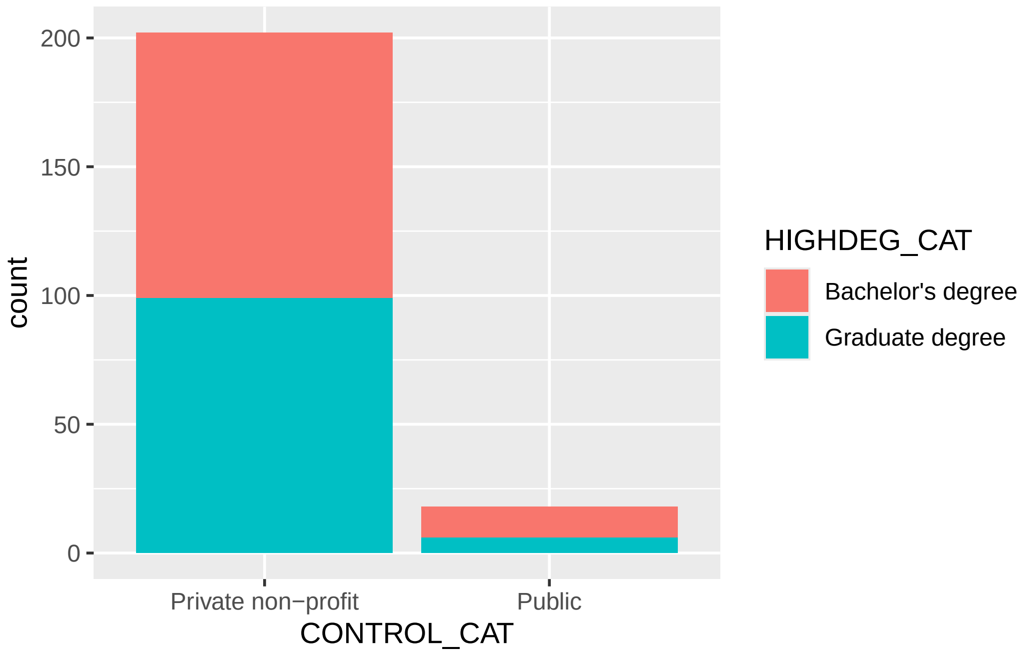
```
# Reorder institution name by the average faculty salaries
pa_schools <- mutate(pa_schools,
                     INSTNM = fct_reorder(INSTNM,
                                          -AVGFACSAL))

# Look at average faculty salaries by school
ggplot(data = pa_schools, mapping = aes(x = INSTNM,
                                        y = AVGFACSAL)) +
  geom_col() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



One more `ggplot` thing: `geom_bar()` versus `geom_col()`

We created the following bar graph that compares the number of public and private schools by highest degree awarded. Notice that `geom_bar()` and `geom_col()` both create bar graphs. When should we use one or the other?

```
# Graph of counts
ggplot(data = colleges, mapping = aes(x = CONTROL_CAT,
                                      fill = HIGHDEG_CAT)) +
  geom_bar()
```



Summarizing data with `dplyr`

`summarize()`: Summarize variable(s)

What is the average admission rate? What is the lowest admission rate?

```
colleges_summary <- summarize(colleges,
                              mean_admit = mean(ADM_RATE, na.rm = TRUE),
                              lowest_admit = min(ADM_RATE, na.rm = TRUE) )
colleges_summary
```

```
# A tibble: 1 x 2
  mean_admit lowest_admit
       <dbl>        <dbl>
1      0.601       0.0693
```

`count()`: Add up number of rows for each category

How many historically black colleges and universities are in the dataset? Of those, how many award graduate degrees?

```r
count(colleges, HBCU)
```

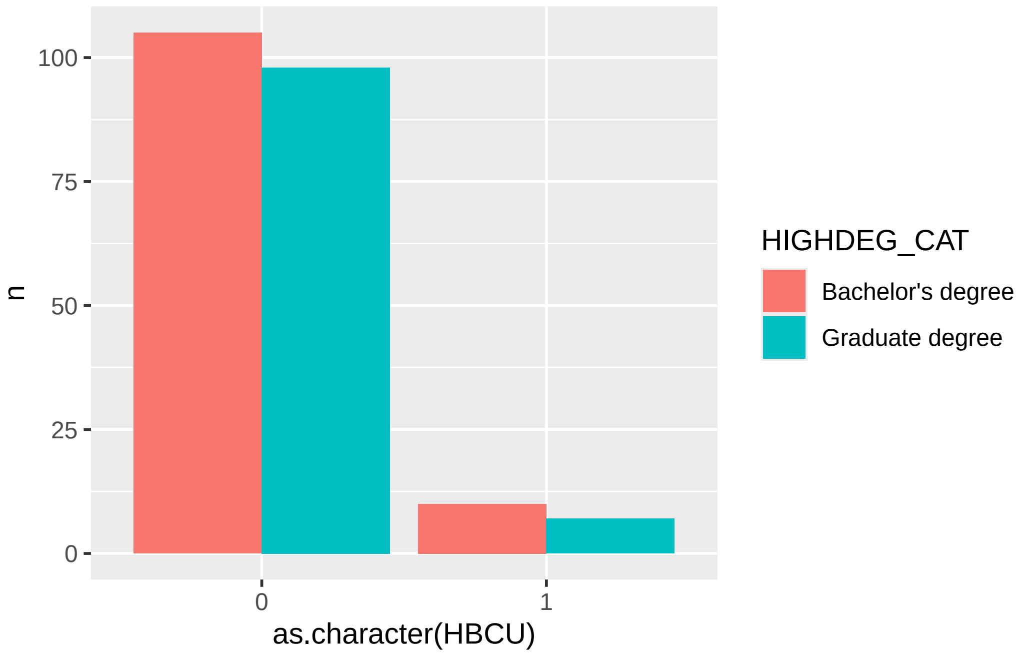
```
# A tibble: 2 x 2
   HBCU     n
  <dbl> <int>
1     0   203
2     1    17
```

```r
count(colleges, HBCU, HIGHDEG_CAT)
```

```
# A tibble: 4 x 3
   HBCU HIGHDEG_CAT           n
  <dbl> <chr>             <int>
1     0 Bachelor's degree   105
2     0 Graduate degree      98
3     1 Bachelor's degree    10
4     1 Graduate degree       7
```

Store the counts of HBCUs and HIGHDEG_CAT as a dataset and then graph it. Should we use geom_col() or geom_bar()?

```r
dat <- count(colleges, HBCU, HIGHDEG_CAT)

ggplot(data = dat, mapping = aes(x = as.character(HBCU), y = n, fill = HIGHDEG_CAT)) +
  geom_col(position = "dodge")
```



Some Thoughts on Wrangling

- Data are messy. Be prepared to wrangle.

  "Tidy datasets are all alike, but every messy dataset is messy in its own way." – Hadley Wickham

- Before you start writing code ask yourself, what do I expect the wrangled data to look like? How many rows do I expect? How many columns?
- Don't try to wrangle all at once.
  - Write one line of code. Run it. And then keep going.
- Give the wrangled dataset a new name if you are removing rows or changing the structure drastically.

Main Data Wrangling Operations in `dplyr`

`summarize()`: Summarize variable(s)

We saw that `summarize()` is good for computing summary statistics.

```r
colleges_summary <- summarize(colleges,
                              mean_admit = mean(ADM_RATE, na.rm = TRUE),
                              lowest_admit = min(ADM_RATE, na.rm = TRUE) )
colleges_summary
```

```
# A tibble: 1 x 2
  mean_admit lowest_admit
       <dbl>        <dbl>
1      0.601       0.0693
```

`count()`: Add up number of rows for each category

`count()` is useful when you have categorical variables and want to know how many observations are in each category.

```r
count(colleges, HBCU)
```

```
# A tibble: 2 x 2
   HBCU     n
  <dbl> <int>
1     0   203
2     1    17
```

```r
count(colleges, HBCU, HIGHDEG)
```

```
# A tibble: 4 x 3
   HBCU HIGHDEG     n
  <dbl>   <dbl> <int>
1     0       3   105
2     0       4    98
3     1       3    10
4     1       4     7
```

`mutate()`: Modify an existing variable or add new variables

Three examples below:

- Adding a new variable called `Location`: indicates if a school is in PA or not.

- Creating `HIGHDEG_CAT`: Which takes the numerical varaible `HIGHDEG` and creates a categorical version.
- Fixing `DEBT_MDN` so that `R` stores it as a numerical variable, not a categorical variable (which `R` calls a character vector).

```r
colleges <- mutate(colleges,
                   Location = if_else(STABBR == "PA", "PA", "NOT PA"),
                   HIGHDEG_CAT = case_match(HIGHDEG,
                                            3 ~ "Bachelor's degree",
                                            4 ~ "Graduate degree"),
                   DEBT_MDN = as.numeric(DEBT_MDN))
#Check work
glimpse(colleges)
```

```
Rows: 220
Columns: 27
$ UNITID          <dbl> 100937, 101912, 106342, 107080, 107512, 112260, 115409~
$ INSTNM          <chr> "Birmingham-Southern College", "Oakwood University", "~
$ CITY            <chr> "Birmingham", "Huntsville", "Batesville", "Conway", "A~
$ STABBR          <chr> "AL", "AL", "AR", "AR", "AR", "CA", "CA", "CA", "CA", ~
$ HIGHDEG         <dbl> 3, 4, 3, 4, 4, 4, 3, 4, 3, 3, 3, 4, 4, 3, 3, 4, 4, 4, ~
$ PREDDEG         <dbl> 3, 3, 3, 3, 3, 3, 3, 1, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, ~
$ CONTROL         <dbl> 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 1, ~
$ HBCU            <dbl> 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
$ TUITFTE         <dbl> 10340, 12279, 10188, 9685, 10749, 35957, 36122, 26331,~
$ AVGFACSAL       <dbl> 7029, 4842, 5817, 7889, 6735, 15333, 14478, 11309, 117~
$ ADM_RATE        <dbl> 0.5717, 0.6805, 0.5984, 0.6028, 0.7232, 0.1035, 0.1336~
$ SATVR75         <dbl> 670, NA, NA, 680, 610, 760, 770, NA, 750, NA, 770, 760~
$ SATMT75         <dbl> 610, NA, NA, 648, 590, 790, 790, NA, 760, NA, 790, 750~
$ ACTCM75         <dbl> 29, NA, NA, 30, 28, 35, 36, NA, 34, NA, 35, 34, NA, 32~
$ COSTT4_A        <dbl> 35495, 38377, 44749, 49928, 43878, 78723, 82236, NA, 7~
$ NPT4_PRIV       <dbl> 19723, 19686, 25183, 22780, 23086, 19489, 39671, NA, 3~
$ UGDS            <dbl> 968, 1378, 489, 1127, 1587, 1383, 906, 15, 1935, 1212,~
$ UG25ABV         <dbl> 0.0170, 0.1284, 0.0276, 0.0054, 0.0140, 0.0021, 0.0011~
$ PCTFLOAN_DCS    <dbl> 0.6452, 0.6477, 0.5934, 0.4483, 0.6109, 0.1627, 0.3646~
$ PCTPELL_DCS     <dbl> 0.2277, 0.4906, 0.3702, 0.2543, 0.2486, 0.2008, 0.1293~
$ DEBT_MDN        <dbl> 16000, 21500, 10699, 19500, 15000, 11948, 19500, 18667~
$ C100_4          <dbl> 0.5854, 0.3351, 0.3085, 0.6743, 0.6174, 0.8318, 0.8826~
$ RET_FT4         <dbl> 0.7746, 0.7706, 0.5072, 0.7905, 0.7897, 0.9579, 0.9733~
$ MD_EARN_WNE_5YR <dbl> 56625, 51429, 45744, 49579, 48168, 108186, 154095, 418~
$ Location        <chr> "NOT PA", "NOT PA", "NOT PA", "NOT PA", "NOT PA", "NOT~
$ CONTROL_CAT     <chr> "Private non-profit", "Private non-profit", "Private n~
```

```
$ HIGHDEG_CAT     <chr> "Bachelor's degree", "Graduate degree", "Bachelor's de~
```

```r
count(colleges, Location)
```

```
# A tibble: 2 x 2
  Location     n
  <chr>    <int>
1 NOT PA     199
2 PA          21
```

```r
count(colleges, HIGHDEG_CAT)
```

```
# A tibble: 2 x 2
  HIGHDEG_CAT           n
  <chr>             <int>
1 Bachelor's degree   115
2 Graduate degree     105
```

`select()`: Extract variables

Let's create a new dataset that only has the school name and location.

```r
colleges2 <- select(colleges, INSTNM, Location)
```

`filter()`: Extract cases

Let's filter down to schools that are:

- In the mid-atlantic: PA, NJ, VA, MD, DE, WV, DC
- Have undergraduate enrollments over 1000 students
- Don't have grad students

```r
colleges3 <- filter(colleges,
                    STABBR %in% c("PA", "NJ", "VA", "MD", "DE", "WV", "DC"),
       UGDS > 1000,
       HIGHDEG != 4)
```

Let's filter down to just Bucknell.

```r
bucknell <- filter(colleges, INSTNM == "Bucknell University")
```

`drop_na()`: Remove rows that have missing values for certain variables

Let's remove rows that are missing an admissions rate.

```r
colleges_adm_rate_complete <- drop_na(colleges, ADM_RATE)
```

More wrangling functions

`arrange()`: Sort the cases

Let's sort rows by their admissions rate. Which schools has the lowest admissions rate? Which has the highest?

```r
arrange(colleges, ADM_RATE)
```

```
# A tibble: 220 x 27
   UNITID INSTNM    CITY  STABBR HIGHDEG PREDDEG CONTROL  HBCU TUITFTE AVGFACSAL
    <dbl> <chr>     <chr> <chr>    <dbl>   <dbl>   <dbl> <dbl>   <dbl>     <dbl>
 1 216287 Swarthmo~ Swar~ PA           3       3       2     0   29620     13487
 2 121345 Pomona C~ Clar~ CA           3       3       2     0   23672     14220
 3 164465 Amherst ~ Amhe~ MA           3       3       2     0   30616     14046
 4 161086 Colby Co~ Wate~ ME           3       3       2     0   34919     11925
 5 168342 Williams~ Will~ MA           4       3       2     0   35531     14484
 6 189097 Barnard ~ New ~ NY           3       3       2     0   42671     14635
 7 161004 Bowdoin ~ Brun~ ME           3       3       2     0   34579     13417
 8 112260 Claremon~ Clar~ CA           4       3       2     0   35957     15333
 9 164155 United S~ Anna~ MD           3       3       1     0       0     12920
10 153384 Grinnell~ Grin~ IA           3       3       2     0   19898     11658
# i 210 more rows
# i 17 more variables: ADM_RATE <dbl>, SATVR75 <dbl>, SATMT75 <dbl>,
#   ACTCM75 <dbl>, COSTT4_A <dbl>, NPT4_PRIV <dbl>, UGDS <dbl>, UG25ABV <dbl>,
#   PCTFLOAN_DCS <dbl>, PCTPELL_DCS <dbl>, DEBT_MDN <dbl>, C100_4 <dbl>,
#   RET_FT4 <dbl>, MD_EARN_WNE_5YR <dbl>, Location <chr>, CONTROL_CAT <chr>,
#   HIGHDEG_CAT <chr>
```

```r
arrange(colleges, desc(ADM_RATE))
```

```
# A tibble: 220 x 27
   UNITID INSTNM    CITY  STABBR HIGHDEG PREDDEG CONTROL  HBCU TUITFTE AVGFACSAL
    <dbl> <chr>     <chr> <chr>    <dbl>   <dbl>   <dbl> <dbl>   <dbl>     <dbl>
 1 172033 Sacred H~ Detr~ MI           4       1       2     0   16565      6336
 2 233611 Southern~ Buen~ VA           3       3       2     0   12302      6313
 3 182917 Magdalen~ Warn~ NH           3       3       2     0   12194      4361
 4 215275 Universi~ Gree~ PA           3       3       1     0   13107      7602
 5 154527 Wartburg~ Wave~ IA           3       3       2     0   16505      6994
 6 233301 Randolph~ Lync~ VA           4       3       2     0   12760      8249
 7 206525 Wittenbe~ Spri~ OH           4       3       2     0   12968      7836
 8 150604 Franklin~ Fran~ IN           4       3       2     0   12805      6801
 9 167288 Massachu~ Nort~ MA           4       3       1     0    6841      9334
10 165936 Gordon C~ Wenh~ MA           4       3       2     0   14956      6992
# i 210 more rows
# i 17 more variables: ADM_RATE <dbl>, SATVR75 <dbl>, SATMT75 <dbl>,
#   ACTCM75 <dbl>, COSTT4_A <dbl>, NPT4_PRIV <dbl>, UGDS <dbl>, UG25ABV <dbl>,
#   PCTFLOAN_DCS <dbl>, PCTPELL_DCS <dbl>, DEBT_MDN <dbl>, C100_4 <dbl>,
#   RET_FT4 <dbl>, MD_EARN_WNE_5YR <dbl>, Location <chr>, CONTROL_CAT <chr>,
#   HIGHDEG_CAT <chr>
```

The pipe: %>% or |> for chaining together multiple wranglings

If you want to do multiple operations at once, you should use the pipe.

Suppose we want to look at `INSTNM`, `Location`, `ADM_RATE`, `UGDS`, `RET_FT4`, and `MD_EARN_WNE_5YR` for schools in PA that reported an admissions rate and we want to arrange the schools from largest undergraduate class to smallest undergraduate class.

```r
PA_colleges <- colleges %>%
  mutate(Location = if_else(STABBR == "PA", "PA", "Not PA")) %>%
  select(INSTNM, Location, ADM_RATE, UGDS, RET_FT4, MD_EARN_WNE_5YR) %>%
  filter(Location == "PA") %>%
  drop_na(ADM_RATE) %>%
  arrange(desc(UGDS))
PA_colleges
```

```
# A tibble: 20 x 6
   INSTNM                        Location ADM_RATE  UGDS RET_FT4 MD_EARN_WNE_5YR
   <chr>                         <chr>       <dbl> <dbl>   <dbl>           <dbl>
```

```
 1 Bucknell University           PA          0.326   3732   0.906             90297
 2 Lafayette College             PA          0.336   2725   0.899             86844
 3 Gettysburg College            PA          0.563   2236   0.886             71373
 4 Susquehanna University        PA          0.767   2139   0.854             59913
 5 Dickinson College             PA          0.349   2083   0.888             71404
 6 Franklin and Marshall College PA          0.362   1986   0.878             68877
 7 Muhlenberg College            PA          0.655   1933   0.909             67290
 8 Swarthmore College            PA          0.0693  1619   0.960             73588
 9 Ursinus College               PA          0.822   1505   0.824             61871
10 Haverford College             PA          0.142   1417   0.961             69576
11 Bryn Mawr College             PA          0.308   1402   0.903             57709
12 Saint Vincent College         PA          0.734   1335   0.84              56756
13 Allegheny College             PA          0.696   1324   0.789             58614
14 University of Pittsburgh-Gre~ PA          0.976   1323   0.633             69754
15 Albright College              PA          0.849   1276   0.640             59794
16 Washington & Jefferson Colle~ PA          0.881   1139   0.829             65052
17 Juniata College               PA          0.762   1116   0.807             53474
18 Lycoming College              PA          0.752   1046   0.721             53116
19 Westminster College           PA          0.753   1023   0.822             53025
20 Bryn Athyn College of the Ne~ PA          0.800    271   0.776             38029
```

`group_by()`: Perform actions by certain groups

For each of the Mid-Atlantic states, what is the average admission rate and how many schools are in each state?

```r
filter(colleges, STABBR %in% c("PA", "NJ", "VA", "MD", "DE", "WV", "DC")) %>%
  drop_na(ADM_RATE) %>%
  group_by(STABBR) %>%
  summarize(mean_admit = mean(ADM_RATE), count = n())
```

```
# A tibble: 5 x 3
  STABBR mean_admit count
  <chr>       <dbl> <int>
1 MD          0.586     5
2 NJ          0.754     2
3 PA          0.595    20
4 VA          0.718    16
5 WV          0.649     1
```

How can I combine two datasets?

- Often the data is stored across several datasets and you want to combine them into one in a principled way.
- Need a key that links the two datasets.

```
# Load data from the Opportunity Insights lab
opportunity_insights <- read_csv("data/opportunity_insights.csv")
```

Suppose we want to add the upward mobility information from the Opportunity Insights dataset to our colleges dataset. Opportunity Insight is a research initiative based at Harvard University and led by Raj Chetty, John Friedman, and Nathaniel Hendren, with the goal of improving upward mobility in the United States by studying barriers to economic opportunity and translating findings into policy change. They defined a college's mobility rate as the percentage of students with parents in the bottom income quintile who ended up in the top x% (in their mid-30s). The variables `mr_kq5_pq1` and `mr_ktop1_pq1` and refer to the percentage of students with parents in the bottom income quintile who ended up in the top 20% and top 1%, respectively.

Let's first look at smaller datasets so we can explore the different types of data joins. What are the key variables?

```
# Create smaller versions
colleges_nyc <- colleges %>%
  select(INSTNM, CITY, STABBR, ADM_RATE) %>%
  filter(CITY == "New York")
colleges_nyc
```

```
# A tibble: 3 x 4
  INSTNM                      CITY     STABBR ADM_RATE
  <chr>                       <chr>    <chr>     <dbl>
1 Barnard College             New York NY       0.0879
2 Marymount Manhattan College New York NY       0.721
3 The King's College          New York NY       0.453
```

```
opp_ny <- opportunity_insights %>%
  filter(state == "NY", tier_name == "Selective private")
opp_ny
```

```
# A tibble: 48 x 5
   name                                  state tier_name mr_kq5_pq1 mr_ktop1_pq1
```

```
   <chr>                                      <chr> <chr>          <dbl>        <dbl>
 1 Adelphi University                         NY    Selectiv~     0.0326   0.00261
 2 Alfred University                          NY    Selectiv~     0.0148   0.0000507
 3 Boricua College                            NY    Selectiv~     0.0364   0.000132
 4 Canisius College                           NY    Selectiv~     0.0236   0.00205
 5 Cazenovia College                          NY    Selectiv~     0.0126   0.000142
 6 Clarkson University                        NY    Selectiv~     0.0297   0.000624
 7 College Of Mount Saint Vincent And M~ NY    Selectiv~     0.0578   0.00173
 8 College Of New Rochelle                    NY    Selectiv~     0.0287   0.00000964
 9 College Of Saint Rose                      NY    Selectiv~     0.0173   0.000686
10 D'Youville College                         NY    Selectiv~     0.0397   0.000104
# i 38 more rows
```

Three common types of joins:

```r
# The inner join
smallest <- inner_join(colleges_nyc, opp_ny, join_by("INSTNM" == "name"))
smallest
```

```
# A tibble: 1 x 8
  INSTNM           CITY  STABBR ADM_RATE state tier_name mr_kq5_pq1 mr_ktop1_pq1
  <chr>            <chr> <chr>     <dbl> <chr> <chr>          <dbl>        <dbl>
1 Marymount Manha~ New ~ NY        0.721 NY    Selectiv~     0.0329     0.000970
```

```r
# The full join
largest <- full_join(colleges_nyc, opp_ny, join_by("INSTNM" == "name"))
largest
```

```
# A tibble: 50 x 8
   INSTNM          CITY  STABBR ADM_RATE state tier_name mr_kq5_pq1 mr_ktop1_pq1
   <chr>           <chr> <chr>     <dbl> <chr> <chr>          <dbl>        <dbl>
 1 Barnard College New ~ NY       0.0879 <NA>  <NA>          NA         NA
 2 Marymount Manh~ New ~ NY       0.721  NY    Selectiv~      0.0329     0.000970
 3 The King's Col~ New ~ NY       0.453  <NA>  <NA>          NA         NA
 4 Adelphi Univer~ <NA>  <NA>    NA      NY    Selectiv~      0.0326     0.00261
 5 Alfred Univers~ <NA>  <NA>    NA      NY    Selectiv~      0.0148     0.0000507
 6 Boricua College <NA>  <NA>    NA      NY    Selectiv~      0.0364     0.000132
 7 Canisius Colle~ <NA>  <NA>    NA      NY    Selectiv~      0.0236     0.00205
 8 Cazenovia Coll~ <NA>  <NA>    NA      NY    Selectiv~      0.0126     0.000142
 9 Clarkson Unive~ <NA>  <NA>    NA      NY    Selectiv~      0.0297     0.000624
10 College Of Mou~ <NA>  <NA>    NA      NY    Selectiv~      0.0578     0.00173
# i 40 more rows
```

```r
# The left join
middle <- left_join(colleges_nyc, opp_ny, join_by("INSTNM" == "name"))
middle
```

```
# A tibble: 3 x 8
  INSTNM           CITY  STABBR ADM_RATE state tier_name mr_kq5_pq1 mr_ktop1_pq1
  <chr>            <chr> <chr>     <dbl> <chr> <chr>          <dbl>        <dbl>
1 Barnard College  New ~ NY       0.0879 <NA>  <NA>          NA          NA
2 Marymount Manha~ New ~ NY       0.721  NY    Selectiv~      0.0329      0.000970
3 The King's Coll~ New ~ NY       0.453  <NA>  <NA>          NA          NA
```

Which join should we use if we want to add the upward mobility information to our colleges dataset?

```r
colleges_plus <- left_join(colleges, opportunity_insights, join_by("INSTNM" == "name"))
```

Resources for Learning More about Data Wrangling with `dplyr`

- Modern Dive's chapter on Data Wrangling
- R for Data Science's chapter on Data Transformation
- `dplyr` cheatsheet: https://raw.githubusercontent.com/rstudio/cheatsheets/main/data-transformation.pdf